

NAF PROJECT & FUNDS TRACKER

AIRTRACK COMMAND SYSTEM

*Comprehensive Project Design, Features, Workflow, Scope Review, and Locked Source-of-Truth
Draft*

Prepared for	Project sponsor and implementation team
Document purpose	Decision-ready design pack to remove out-of-scope items, add missing items, and lock the official baseline
Deployment model	Desktop-first, offline-first, centrally synced command platform
Draft status	Working source-of-truth version for review and lock-down

NEW UPDATE

The system will have only clerk desktop app version which can always synchronise to online server. Others including the CAS will assess their accounts online. A lite mobile app not more than 5M will be used for alerting Tasked branches and their AAs/staff officers which they will in turn go online to do tasks in their dashboard. Same also to CAB. This rule overrides other related rules

This document consolidates the project concept, feature set, workflow, architecture, data direction, roles, windows, and review checklist into one clean blueprint. It is designed so you can strike out what is not needed, insert what is missing, and then declare one locked source of truth for the whole team.

1. Executive Summary

The NAF Project & Funds Tracker, also called AirTrack Command System, is a memo-to-execution tracking platform for recording financial request letters (memo) upon approval by the Chief of the Air Staff (CAS) and following them up (tracking), fund release, commencement, progress reporting, validation, completion, and archival.

The system is not just a register. It is a command workflow platform with search, accountability, analytics, alerts, audit trails, export tools, and desktop clients for assigned branches and Direct-to-CAS offices.

The recommended implementation model is desktop-first and offline-first. Each client installation stores work locally, then synchronizes to a central online database whenever internet is available or when the user clicks Sync Now. The system will have a lite cross platform mobile app that is linked to each client user computer to be operated by the AA to Branch Chief or His clerk, the lite app will be for notification of tasks which will be installed on mobile phones of the Branch Chief to put the clerk or AA in check.

2. Project Objectives

- Create one authoritative record of approved, rejected, and KIV financial memos.
- Track the full lifecycle of approved memos from fund release to completion and archival.
- Assign clear follow-up responsibility through Primary Monitor and Final Validator roles.
- Provide CAS and HQ leadership with fast search, modern dashboards, alerts, and analysis of how funds are requested, released, and utilized.
- Support offline desktop usage across branches, with reliable synchronization to a central server.
- Preserve audit history for every major action and provide controlled locking to prevent tampering.

3. In-Scope Modules

Module	Purpose	Core Outcomes
Memo Registry	Register financial request memos and attachments	Draft, KIV, Rejected, Approved records
Assignment & Tracking	Assign Primary Monitor and Final Validator	Ownership, deadlines, roadmap
Fund Release Desk	CAB queue for release decision	Rejected, Partial, Full fund release
Commencement Feedback	Confirm whether funded work has started	Commenced, Not Commenced, comments
Progress Reporting	Periodic updates from monitoring branches	Progress history, evidence, delays
Validation Desk	Final completion verification	Completed or Validation Rejected
Analytics & Reporting	CAS dashboard and management reports	Charts, trends, filters, exports
Archive & Recovery	Preserve closed records safely	Read-only archive with restore control

Offline Desktop Client	Fast light client for branch users	Local work, reminders, sync
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4. Out-of-Scope Candidates for Review

These are intentionally listed for decision. Keep only what clearly serves the command workflow and budget reality.

- ☐ Native mobile apps in version 1. The desktop-first client already covers the operational need.
- ☐ Very advanced AI forecasting before the core workflow is stable.

5. Missing-Items Review Checklist

- ☐ Should all 36 states plus FCT be maintained in master data for project location reporting? yes
- ☐ Should branch-level dashboards differ by role, or should all operational users share one simplified desktop experience? **Yes operational users share one simplified desktop experience, only the branch which is required during registration appears on the dashboard as unique and only sees and keep record of what is sent to the branch only**
- ☐ Should approval authority be strictly read-only for CAS Viewer while another internal authority role performs approval actions?
- ☐ Should documentary-completion categories auto-complete immediately after full fund release, or pass through a lighter confirmation step? **Yes, Auto complete**
- ☐ Should attachments support only documents and images, or also short video/audio evidence? **Only documents and images**
- ☐ What are the final retention rules before a completed record moves to archive? **In the archive table auto listed as per date and sortable**

6. Standardized Category Headings and Tracking Modes

No.	Category Heading	Tracking Mode	Typical Behavior
1	Projects & Infrastructure	FULL	Full lifecycle, physical execution, validation required
2	Procurement, Logistics & Supply	FULL	Delivery, inspection, receiving evidence
3	Training, Courses & Development	CONTROLLED	Documentary completion can be lighter
4	Operations & Mission Support	FULL	Operational progress and evidence trail

5	Finance, Budget & Payments	CONTROLLED	Payment confirmation focused
6	Travel, Allowances & Foreign Assignments	CONTROLLED	Travel and payment completion focused
7	Medical, Welfare & Human Support	CONTROLLED	Support delivery confirmation focused
8	Personnel & Administration	CONTROLLED	Administrative completion focused
9	Maintenance & Engineering Services	FULL	Technical progress and completion evidence
10	Contracts, Services & General Support	FULL	Deliverable and contract stage tracking

7. Role Design

Role	Main Permissions	Restrictions / Notes
CAS Viewer	Read-only strategic view, charts, alerts, newest completed items and can sort from beginning to end as deemed fit	No operational editing
Super Admin (PASO)	Full access, users, branches, settings, backups, restore, controlled override, sees branch not doing their job and query them in app, can create disable and delete any user. Should show project tracked level(delivered/read by tracker or validators using sing tick and double tick green)	Should be tightly controlled
Clerk	Create memo, edit before lock, upload documents, assign monitor and validator where permitted, sees branch not doing their job and report to super admin(PASO)	Cannot falsely validate execution

CAB Officer / CAB Desk	Receive release queue, reject, partially release, fully release	CAB section locks at 100% release
Primary Monitor	Confirm commencement, submit periodic progress, upload evidence	Only for assigned records
Final Validator	Approve or reject completion with reason	Required for FULL tracking
Client Branch / DRU User	Respond to assigned actions and alerts	Limited to role-allowed tasks

8. Official Workflow

Stage 1. Memo Registration

Clerk records memo reference, heading, branch/DRU, beneficiary, amount, currency, date, approval state, notes, **Tel and attachment (optional)**.

Stage 2. Assignment

If approved, the system assigns Primary Monitor and Final Validator, creates the roadmap, and moves the record to Awaiting Fund Release **page**.

Stage 3. Fund Release Desk

CAB receives a queue, chooses release rejection, partial release **(require next release date, to notify)**, or full release, and the system stamps officer, amount, percentage, and time.

Stage 4. Commencement Feedback

After funds are released, the assigned monitoring branch confirms whether execution has commenced within the configured reminder period.

Stage 5. Periodic Progress

Once underway, the Primary Monitor submits progress, achievements, delays, next action, evidence, and percentage updates every 15 days.

Stage 6. Validation

When progress reaches 100%, the Final Validator approves completion or rejects it with a mandatory reason (CAS also sees this).

Stage 7. Completion and Archive

Completed items **appears with full details in completed project page in leadership views**, remain searchable with linked evidence, and later move to archive under retention rules **in the appropriate page which can be viewed by the CAS**.

9. Official Lifecycle Statuses

Lifecycle Status	Used In	Meaning
Draft	Registry / CAB / Monitor / Validator	Record is being prepared and can still be edited or deleted by Clerk.

KIV (This status cannot be sent to CAB until changed to approved)	Registry / CAB / Monitor / Validator	Memo is kept in view and not progressing operationally.
Rejected	Registry / CAB / Monitor / Validator	Memo has been rejected and is preserved for reference in the rejected memos page
Approved	Registry / CAB / Monitor / Validator	Memo is approved and can move to assignment and fund release workflow.
Awaiting Fund Release	Registry / CAB / Monitor / Validator	Approved record is pending CAB action. CAS and PASO can view this in the pending funds release page
Fund Release Rejected	Registry / CAB / Monitor / Validator	CAB rejected release and recorded reason.
Partially Funded	Registry / CAB / Monitor / Validator	CAB released less than 100%; reminders continue.
Funds Released	Registry / CAB / Monitor / Validator	Fund release completed; downstream execution workflow begins.
Awaiting Commencement Feedback	Registry / CAB / Monitor / Validator	System is waiting for commencement response.
Not Commenced	Registry / CAB / Monitor / Validator	Assigned branch reported that work has not started.
Ongoing	Registry / CAB / Monitor / Validator	Execution is underway and periodic progress continues.
Awaiting Validation	Registry / CAB / Monitor / Validator	Primary Monitor reported 100% and validator review is pending.
Validation Rejected	Registry / CAB / Monitor / Validator	Validator rejected the completion claim with reason.
Completed	Registry / CAB / Monitor / Validator	Completion has been accepted and posted in the relevant page viewable by the CAS, PASO and

can be sorted as completed
project in the CAS Dashboard.

Archived

Registry / CAB / Monitor /
Validator

Closed read-only record,
restorable only by authorized
flow.

10. Desktop-First Technical Architecture

Recommended model: Electron desktop client + local SQLite database + central Node.js API + central PostgreSQL database. The desktop app works offline, queues changes locally, and synchronizes automatically when the internet returns or when the user clicks Sync Now.

Layer	Recommended Design
Client	Electron desktop app with role-aware UI and local SQLite storage
Offline data	SQLite local store with sync queue and conflict-safe identifiers
Server API	Node.js backend enforcing lifecycle rules, permissions, audit, and exports
Central database	PostgreSQL for master records, reporting, search, and archive
File storage	Structured document and evidence storage with attachment metadata
Sync	Automatic online detection plus manual Sync Now button
Security	Role-based access, password hashing, device/session tracking, audit logs, HTTPS

11. Hosting Recommendation

Because the data size is expected to remain small over many years, the system does not need a heavy or expensive infrastructure stack. A managed PostgreSQL deployment is the most practical starting point for production, while keeping the architecture portable so the database can later move to a private server if policy requires it.

Option	Strength	Trade-off	Suggested Use
Managed PostgreSQL	Easy administration, backup convenience, fast launch	Monthly cost	Preferred production starting point
Small secured VPS	More control, potentially cheaper long-term	More admin overhead and security responsibility	Later migration if tighter control is required

Free cloud tiers

Low initial cost

Not suitable for mission-critical production use

Development and pilot only

12. Core Data Design

The backend should be built around a memo-centered data model. Supporting entities include categories, branches/DRUs, roles, assignments, fund release events, commencement feedback, progress reports, validation reviews, attachments, alerts, audit logs, backups, exports, sync devices, and sync queue logs.

Entity	Purpose
memos	Main registry record
memo_assignments	Primary Monitor and Final Validator assignment
fund_release_events	CAB actions and locking record
commencement_feedback	Has work started or not
progress_reports	Periodic updates and evidence trail
validation_reviews	Completion approval or rejection
attachments	Linked documents, photos,
audit_logs	Who changed what, when, and why
notifications	Alerts and reminders
sync_devices / sync_queue_logs	Desktop sync traceability

13. Memo Registration Form Design

The form should feel like a proper operational form, not a spreadsheet disguised as a screen. Required core fields include Memo Reference Number, Memo Description, Category Heading, Tracking Mode, Originating Branch or DRU, Beneficiary Type, Beneficiary Name + Tel No, State(**sortable by CAS**), Amount, Currency (**sortable by CAS**), Approval Status, Approval Date, Attachment, and Remarks.

Conditional fields should open when relevant. If the memo is approved, show Primary Monitor, Final Validator, Expected Start Date, Expected End Date, and Fund Release Required. If the selected heading is controlled and does not need physical execution, validation can be optional depending on final policy.

14. Category-Specific Extra Fields

Category	Recommended Extra Fields
Projects & Infrastructure	Project Location (State), Contractor Name, Start Date, End Date, Physical Progress %, Financial Progress %, Site Photos
Procurement, Logistics & Supply	Supplier / Vendor, Item Type, Quantity, Delivery Date, Delivery Status, Inspection Status, Receiving Officer, Store Entry Reference
Training, Courses & Development	Training Type, Beneficiary, Institution, Number of Participants, Start Date, End Date, Attendance Confirmed Require CAB Release of funds(Validator optional)
Operations & Mission Support	Operation Name, Operation Location, Start Window, End Window, Supporting Unit, Delivery / Mission Status Require CAB Release of funds(Validator optional)
Finance, Budget & Payments	Payment Type, Beneficiary Class, Due Date, Payment Completion Status
Travel, Allowances & Foreign Assignments	Travel Type, Destination, Ticket Status, Visa Status, Departure Date, Return Date Require CAB Release of funds(Validator optional)
Medical, Welfare & Human Support	Support Type, Hospital / Institution, Beneficiary, Treatment / Welfare Status Require CAB Release of funds(Validator optional)
Personnel & Administration	Action Type, Personnel Count, Committee / Board Reference, Administrative Completion Status
Maintenance & Engineering Services	Equipment / Platform, Maintenance Type, Planned Date, Completion Date, Serviceability Status, Technical Report Upload Require CAB Release of funds(Validator optional)
Contracts, Services & General Support	Contract Reference, Service Provider, Contract Amount, Contract Stage, Deliverable Confirmation Require CAB Release of funds(Validator optional)

15. Alerts and Reminder Rules

Alert Area	Rule
CAB Alerts	Blinking and countdown-based reminder after approval; if partial release remains unresolved, persistent reminder continues.
Commencement Alerts	After funds release, the assigned branch receives a reminder to confirm commencement within the configured period.
Progress Alerts	Every 15 days until completion; overdue items should surface in both client and CAS views.
Validation Alerts	When progress reaches 100%, the Final Validator receives immediate notification.
CAS Alerts	Newest completed items, overdue items, blocked items, partially funded items, and rejected validation cases.

16. Dashboard Design Standard

The CAS dashboard should surface the command picture quickly. Recommended widgets include Total Records, Total Approved, Total KIV, Total Rejected, Total Funds Requested, Total Funds Released, Total Ongoing, Total Completed, Total Overdue, Pending Validation, Pending CAB Action, Newest 5 Completed, and Projects by States. Charts should include Total Funds by Branch / DRU, Ongoing vs Completed, Approved vs KIV vs Rejected, Monthly / Quarterly / Yearly trends, Category distribution, Currency distribution, and custom date-range analysis and custom keyword search for particular projects to return full details

17. Record Locking Rules

- Draft can be edited or deleted by Clerk.
- Approved records cannot be deleted without Super Admin authority.
- Once CAB sets 100% fund release, the CAB section locks against further CAB edits.
- Completed and validated records cannot be edited except through controlled Super Admin override.
- Archived records are read-only except through authorized restore flow.

18. Audit, Security, and Accountability

Every major action must log user, role, branch or DRU, action taken, previous value, new value, timestamp, and comment or reason where applicable. The system must log create, edit, delete, restore, approve, assign monitor, assign validator, release funds, reject release, commencement response, progress update, validation decision, and archive.

Security design should include hashed passwords, role-based access control, session and device awareness, attachment access control, secure synchronization, and export logging.

19. Division of Labour and Windows

Window	Focus	Suggested Team	Main Deliverables
IW	Integrator	1	Architecture, source of truth, coordination
A	Backend & API	2	APIs, lifecycle guards, auth, audit, exports
B	Database & Sync	2	Schema, seeds, SQLite mirror, sync design
C	Web Dashboard	2	CAS, CAB, validation, archive, analytics UI
D	Desktop Client	2	Offline client, reminders, sync, assigned actions
E	UI/UX & Forms	1-2	Dynamic forms, roadmap, interaction logic
F	Notifications & Automation	1	CAB, commencement, progress, validation alerts
G	Security & Audit	1	Permissions, session/device rules, hardening
H	QA & Validation	1-2	Functional, role, sync, and lifecycle testing

20. Recommended Development Sequence

1. Integrator defines and freezes the official source of truth.
2. Window B produces schema, seeds, ERD, and sync mirror.
3. Window A produces API contract and lifecycle rules.
4. Window E finalizes form, dynamic field logic, and UX decisions.
5. Window C builds the web dashboards and registry.
6. Window D builds the offline desktop client.

7. Window F wires reminders and automation.
8. Window G performs security hardening and audit assurance.
9. Window H runs end-to-end QA.
10. Integrator approves release and manages controlled change requests.

21. Branch and Direct-to-CAS Master Data

HQ branches to preload: Policy and Plans Branch; Training Branch; Operations Branch; Air Intelligence Branch; Aircraft Engineering Branch; Communication and Information Systems Branch; Logistics Branch; Administration Branch; Air Secretary Branch; Accounts and Budget Branch; Medical Services Branch; Transformation and Innovation Branch; Civil Military Relation Branch.

Direct-to-CAS list currently identified: PIMT; BOSE; Directorate of Air Police; Directorate of Procurement; Directorate of Public Relations & Info. Any remaining offices should be confirmed and seeded before the final lock.

22. Final Review Matrix Before Locking the Source of Truth

Review Area	Decision Needed	Final Choice
App identity	Confirm final official product name and abbreviation	CAS Command Funds & Projects System (CFPS)
Approval authority	Confirm which role performs approval actions	Approval is a status selected by clerk on creating record
Controlled categories	Confirm whether they auto-complete at fund release or follow light confirmation	Yes auto-complete at fund release
Attachment policy	Confirm allowed file types and size limits	3MB
Archive policy	Confirm retention timeline before archive	5mins
Reminder timing	Confirm countdown and escalation timings	immediately
Direct-to-CAS offices	Confirm final master list	PASO
Hosting model	Confirm managed PostgreSQL or private VPS	Postgre SQL and later private server